Yissum Research Intelligence Report — Healthcare — Week 32, 2026

5 high-potential paper(s) · Healthcare · Weekly · Week 32, 2026 · Generated August 03, 2026

Hebrew University researchers are advancing precision medicine with groundbreaking single-cell sequencing technologies and AI platforms that reveal hidden disease mechanisms and accelerate biomarker discovery. Concurrently, new therapeutic avenues are emerging, including innovative non-opioid pain relief compounds and smart biomaterials designed for targeted drug delivery. Additionally, novel molecular targets have been identified for precision pharmacological interventions in neurological disorders. These innovations collectively offer high-impact opportunities across diagnostics, drug development, and advanced materials.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Oren Ram** — Genomics, Diagnostics, Drug Discovery, Clinical
- **Avi Priel** — Drug Discovery, Neuroscience, Clinical
- **Itamar Willner** — Synthetic Biology, Drug Discovery, Materials, Diagnostics
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics
- **Mor Nitzan** — Drug Discovery, Software/AI, Genomics, Immunology

HEALTHCARE — RESEARCH HIGHLIGHTS

1. sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells. 35/50

"New single-cell sequencing platform offers cheaper, comprehensive RNA analysis, revolutionizing precision diagnostics."

Main researcher: Oren Ram

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, The Hebrew University, 9190401 Jerusalem, Israel.

Published in Nucleic acids research · 2026-04

ABOUT THIS RESEARCH

Researchers have developed a new single-cell sequencing technology called sc-rDSeq that captures a much wider range of RNA types, including non-polyadenylated RNAs, at a lower cost and higher density than current standards. By applying this to lung cancer, they demonstrated the ability to identify hidden cell subpopulations and drug-resistance mechanisms that traditional methods miss.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

COMMERCIALISATION METRICS

Novelty 8/10

The technology represents a significant platform innovation by enabling full-length, non-polyA RNA capture in a scalable droplet format, overcoming a major technical limitation of current industry standards like 10x Genomics. This capability to access the 'dark transcriptome' (ncRNA, histone RNA, etc.) provides a new dimension of biological insight that is highly relevant for precision oncology.

Commercial Potential 8/10

The technology offers a significant, platform-enabling improvement over industry standards (10x Genomics) by capturing non-polyA RNA and providing higher UMI density, which has clear commercial utility for drug discovery and precision oncology. Its scalability and cost-effectiveness position it as a highly licensable tool for large-scale clinical biomarker profiling.

Market Size 8/10

The technology targets the high-value single-cell sequencing market by offering a platform-enabling solution for comprehensive transcriptomics, which is essential for advanced drug discovery and personalized oncology. Its ability to capture non-polyA RNAs and alternative splicing at scale provides a significant competitive edge in the expanding precision medicine and biomarker discovery sectors.

Tech Readiness 4/10

The technology is currently at the proof-of-concept stage, demonstrated through successful validation in a laboratory setting using cancer cell lines. While it shows significant performance improvements over existing methods, it has not yet been transitioned into a standardized, commercially available kit or integrated into clinical diagnostic workflows.

IP Strength 7/10

The technology represents a potentially patentable platform innovation by optimizing droplet-based capture for non-polyA RNA, offering a clear technical advantage over industry standards like 10x Genomics. However, defensibility may be challenged by the crowded landscape of single-cell sequencing chemistries and the difficulty of protecting specific microfluidic or bead-based optimizations once widely adopted.

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2. Carbamate-based cannabinoids uncover a conserved lipid-sensing pocket in nociceptive TRPA1 and TRPV1 channels. 33/50

"Novel cannabinoid derivatives show high potency, paving the way for non-opioid chronic pain treatments."

Main researcher: Avi Priel · avi.priel@mail.huji.ac.il.

The Institute for Drug Research (IDR), Department of Pharmacology, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Israel. Electronic address: avi.priel@mail.huji.ac.il.

Published in Pharmacological research · 2026-06

ABOUT THIS RESEARCH

Researchers have engineered new cannabinoid-based molecules that more effectively target TRPA1 and TRPV1 ion channels by adding a specific carbamate functional group. These modified compounds show significantly higher potency in activating these pain-sensing channels compared to natural CBD or CBG.

COMMERCIAL ANGLE

The development of these high-potency carbamate derivatives provides a structural roadmap for designing a new class of non-opioid analgesics for chronic pain management.

COMMERCIALISATION METRICS

Novelty 7/10

The research demonstrates significant chemical innovation by using rational design to transform known scaffolds into potent dual agonists, revealing a previously unknown conserved lipid-sensing mechanism. While it offers high target-validation value and potential for new drug classes, it builds incrementally on established phytocannabinoid chemistry rather than introducing an entirely new paradigm.

Commercial Potential 7/10

The research demonstrates successful rational design of small-molecule derivatives with enhanced potency and selectivity, representing a clear path toward a novel non-opioid analgesic lead series. However, the transition from structural discovery to a proprietary drug candidate requires significant validation of pharmacokinetic profiles and safety to reach high-tier commercial readiness.

Market Size 9/10

The research targets TRPA1 and TRPV1, two of the most high-value pathways in the multi-billion dollar non-opioid analgesic market. By developing potent dual-agonists, this work addresses a massive unmet medical need for chronic and acute pain management with significant commercial scalability.

Tech Readiness 3/10

The research is currently at the basic discovery/proof-of-concept stage, utilizing in vitro heterologous expression systems to validate molecular binding and potency. It lacks in vivo animal models, pharmacokinetic data, or toxicity profiling required to advance toward preclinical development.

IP Strength 7/10

The research demonstrates successful rational ligand design to create novel, highly potent carbamate-functionalized derivatives with improved selectivity, providing a clear pathway for composition-of-matter patent claims. While the chemical scaffolds are derivative of known phytocannabinoids, the specific structural modifications and the discovery of a conserved binding pocket offer significant defensibility for new non-opioid analgesic candidates.

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3. Dynamic Switchable and Transient DNA Condensates Driven by Aptamer-Ligand or Ion-Nucleobase Bridged Complexes.

28/50

"Programmable DNA nanocarriers enable smart, triggered drug delivery and responsive biomaterials."

Main researcher: Itamar Willner

The Institute of Chemistry, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published in Small (Weinheim an der Bergstrasse, Germany) · 2026-06

ABOUT THIS RESEARCH

Researchers have developed a way to create DNA-based liquid droplets that can be programmed to form, dissolve, or change their properties on command using specific chemical triggers. These programmable 'coacervates' act as smart materials that can respond dynamically to their environment through molecular switching.

COMMERCIAL ANGLE

This technology enables the development of highly responsive, programmable drug delivery vehicles and smart biomaterials that release payloads only in response to specific biological or chemical signals.

COMMERCIALISATION METRICS

Novelty 8/10

The research introduces a highly innovative approach to controlling DNA phase separation through programmable, stimuli-responsive molecular bridges (aptamer-ligand and metal-ion/mismatch), moving beyond static coacervates toward dynamic, switchable materials. This represents a platform-enabling advancement for responsive drug delivery and programmable soft matter.

Commercial Potential 5/10

The technology demonstrates sophisticated fundamental control over DNA matter, but the abstract lacks specific application towards a high-value end-market such as targeted drug delivery or programmable diagnostics. While it serves as a platform-enabling material, the path to a specific, licensable product remains academically abstract rather than industrially defined.

Market Size 5/10

While the platform offers high scientific novelty for programmable materials, the specific application remains a fundamental research tool rather than a clearly defined high-value commercial market. The market size is currently constrained by its niche status in synthetic biology and materials science without a clear immediate path to large-scale therapeutic or industrial adoption.

Tech Readiness 3/10

The work describes fundamental proof-of-concept molecular design and basic biochemical behavior of DNA-based phase separation. It is currently at the basic research/early laboratory stage without evidence of integrated system testing or biological application scaling.

IP Strength 7/10

The use of programmable DNA architecture and specific chemical bridging (metal-ions/mismatched bases) provides a strong foundation for composition-of-matter patents. However, the inherent 'openness' of DNA nanotechnology and the high likelihood of prior art in DNA condensation and aptamer chemistry may complicate broad claim construction.

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4. Phosphoproteomic profiling reveals reversal of dysregulated kinase signaling by nitric oxide inhibition in the Shank3 mouse model of autism. 28/50

"Key molecular targets identified for precision pharmacological intervention in autism spectrum disorders."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

Institute for Drug Research, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. Haitham.amal@mail.huji.ac.il.; The Rosamund Stone Zander and Hansjoerg Wyss Translational Neuroscience Center, Boston Children's Hospital, Boston, MA, USA. Haitham.amal@mail.huji.ac.il.

Published in Scientific reports · 2026-06

ABOUT THIS RESEARCH

Researchers used phosphoproteomic profiling to map how nitric oxide inhibition restores normal signaling pathways in a mouse model of autism related to the SHANK3 gene. The study identifies specific protein phosphorylation changes that could explain how targeting nitric oxide synthase can reverse behavioral and synaptic symptoms.

COMMERCIAL ANGLE

The findings provide high-resolution molecular targets for the development of precision pharmacological interventions aimed at correcting dysregulated kinase signaling in autism spectrum disorders.

COMMERCIALISATION METRICS

Novelty 6/10

The study provides meaningful mechanistic insight by mapping the signaling pathways downstream of NO inhibition, but it follows a predictable biological logic rather than introducing a groundbreaking new paradigm or platform technology.

Commercial Potential 6/10

The study identifies a mechanistic target (nNOS inhibition) for a high-unmet-need indication, but the phosphoproteomic focus is primarily descriptive rather than proposing a novel, druggable modality or proprietary platform.

Market Size 9/10

Autism spectrum disorder represents a massive, global therapeutic market with significant unmet medical need and a high prevalence across diverse demographics. Targeting a fundamental neurobiological pathway like nNOS/phosphoproteomics offers potential for high-value, broad-spectrum clinical application.

Tech Readiness 3/10

The research is currently at the fundamental discovery stage, utilizing animal models (Shank3 mice) to identify mechanistic signaling pathways rather than testing a clinical candidate in humans. While it provides high-value biological validation, it lacks the preclinical development or pharmacological optimization required for higher TRL scores.

IP Strength 4/10

The paper describes fundamental mechanism-of-action (MoA) discovery rather than a novel composition of matter; while identifying signaling pathways is valuable for target validation, the use of nitric oxide inhibition is an established biological phenomenon that offers limited new patentable space.

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5. Identifying maximally informative signal-aware representations of single-cell data using the information bottleneck. 28/50

"AI framework distills complex single-cell data, accelerating biomarker discovery for drug development and precision medicine."

Main researcher: Mor Nitzan · mor.nitzan@mail.huji.ac.il.

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel; Racah Institute of Physics, The Hebrew University of Jerusalem, Jerusalem, Israel; Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. Electronic address: mor.nitzan@mail.huji.ac.il.

Published in Cell systems · 2026-05

ABOUT THIS RESEARCH

This paper introduces bioIB, an AI framework that uses the information bottleneck method to distill complex single-cell RNA sequencing data into highly specific, interpretable gene clusters. By filtering out noise and focusing on relevant biological signals like disease states, it maps out the hierarchical relationships between different cell types and biological processes.

COMMERCIAL ANGLE

The framework can be commercialized as a high-throughput software platform for drug discovery and precision medicine to identify specific cellular biomarkers and disease drivers from noisy single-cell datasets.

COMMERCIALISATION METRICS

Novelty 6/10

The paper applies an established information theory framework (Information Bottleneck) to a well-trodden domain (scRNA-seq), representing an incremental methodological improvement rather than a fundamental scientific breakthrough. While the application to gene clustering and hierarchical structure is useful, it follows existing trends in deep learning-based dimensionality reduction and feature selection.

Commercial Potential 6/10

The framework presents a strong platform-enabling computational tool for drug discovery and biomarker identification, but its commercial value is currently tied to software/algorithm licensing rather than a standalone proprietary product. Its utility depends on integration into existing bioinformatics pipelines or single-cell analysis platforms.

Market Size 8/10

The framework acts as a platform-enabling technology for the massive and growing single-cell multi-omics market, offering high value for drug discovery and precision medicine. By optimizing signal extraction in complex datasets, it addresses a critical bottleneck in the multi-billion dollar bioinformatics and clinical diagnostics sectors.

Tech Readiness 4/10

The paper presents a computational framework and algorithmic methodology (bioIB) rather than a physical product or clinical tool. While it demonstrates utility on existing datasets, it remains at the fundamental software/algorithm development stage (TRL 3-4) rather than a validated industrial platform.

IP Strength 4/10

The innovation is primarily a computational framework based on the Information Bottleneck method, which is an established theoretical principle in machine learning, making the core algorithm difficult to patent. While the application to scRNA-seq is valuable, the defensibility is low as it likely falls under 'mathematical methods' and is susceptible to rapid replication by open-source community developments.

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